

Daiwei Chen

daiwei.chen@wisc.edu | +1 445-208-7806 | Website | LinkedIn | Google Scholar

RESEARCH FOCUS: GENERATIVE RECOMMENDATION & PERSONALIZATION

Ph.D. researcher working on generative retrieval, Semantic-ID representations, and learning individual preferences. First-author GTI (EMNLP 2026) and PAL (ICLR 2025), with LinkedIn GenAI research experience.

EDUCATION

University of Wisconsin-Madison

Ph.D. in Electrical and Computer Engineering

Advisor: Ramya Korlakai Vinayak

Madison, WI

Sep 2023 – Present

University of Pennsylvania

M.S. in Systems Engineering

Philadelphia, PA

Aug 2021 – May 2023

Zhejiang University

B.S. in Psychology

Hangzhou, China

Sep 2017 – Jun 2021

INDUSTRY RESEARCH EXPERIENCE

LinkedIn

Mountain View, CA

Responsible AI - Research Intern

Jun 2026 – Sep 2026

- Developed **Multi-Teacher On-Policy Distillation (MOPD)** to address the safety-capability trade-off in LLM post-training.

Generative AI - Research Intern

May 2025 – Aug 2025

- Developed **GTI** for LLM-based candidate retrieval; achieved a **21.6% relative Precision@5 gain** over vanilla SFT in offline industrial evaluation (Good Match criterion).
- Introduced a grounding stage that learns new **Semantic-ID embeddings** from paired text while freezing the LM backbone, followed by fine-tuning for generative retrieval.

SELECTED RESEARCH & PUBLICATIONS

Grounded Token Initialization for New Vocabulary in LMs for Generative Recommendation

D. Chen, Z. Fu, C. Jiang, H. Zhang, R. Zhou, T. Wang, C. Yao, G. Li, R. Cai, Y. Cao, R. Jiang, F. Borisyuk, J. Shen, J. Wu, R. K. Vinayak

EMNLP 2026 Main Conference

Grounding Semantic-ID embeddings in pretrained LMs improves generative retrieval and recommendation.

LinkedOut: Linking World Knowledge Representation Out of Video LLM for Next-Generation Video Recommendation

H. Zhang, Y. Lu, L. Wang, Y. Li, D. Chen, Y. Xu, Y. Fu

Findings of CVPR 2026

Fusing world-knowledge representations from video LLMs enables personalized recommendation from raw frames.

PAL: Sample-Efficient Personalized Reward Modeling for Pluralistic Alignment

D. Chen, Y. Chen, A. Rege, Z. Wang, R. K. Vinayak

ICLR 2025 | NeurIPS 2024 Workshop (**Spotlight**) | ICML 2024 Workshop (**Oral**)

Shared preference prototypes enable sample-efficient personalization across text and image preference tasks.

One Model, Many Modes: Evolving System Prompt Ensembles for Creative Generation

D. Chen, A. He, B. Yao, J. Hu, R. K. Vinayak

Under review

Evolved system-prompt ensembles improve novelty and diversity without retraining or sacrificing quality.

Compositional Self-Improvement

C. Shin, D. Chen, J. Cooper, B. Lake, F. Sala, R. K. Vinayak

ICML 2026 – 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents

Composing reliable subproblem predictions enables self-training beyond the original supervision regime.

SERVICE & AWARDS

Reviewer: NeurIPS 2026, ICLR 2026, CVPR 2026, EACL 2026, COLM 2026.

Awards: Lambda.ai Research Grant Sponsorship (2025); UPenn Outstanding Research Award (2023).